

Practical-6

Rendering External JSON Data using Fetch API, Search and Filter

Theory

1. How is JSON fetched, parsed, and rendered?

Answer: JSON is fetched using the Fetch API. First, JavaScript uses `fetch()` to get the JSON file. Then the response is converted into JavaScript data using `.json()`. After parsing, the data is used to create HTML elements and display the information on the webpage.

So, the basic process is:

Fetch JSON → Parse JSON → Use Data → Render on Webpage

2. Which array methods are used for search, filter, sort, and pagination?

Answer: For searching, filtering, sorting, and pagination, different JavaScript array methods are used.

- **Search:** `find()` or `findIndex()`
- **Filter:** `filter()`
- **Sort:** `sort()`
- **Pagination:** `slice()`

These methods help to easily manage and display JSON data on the webpage.

3. How are loading and error states handled?

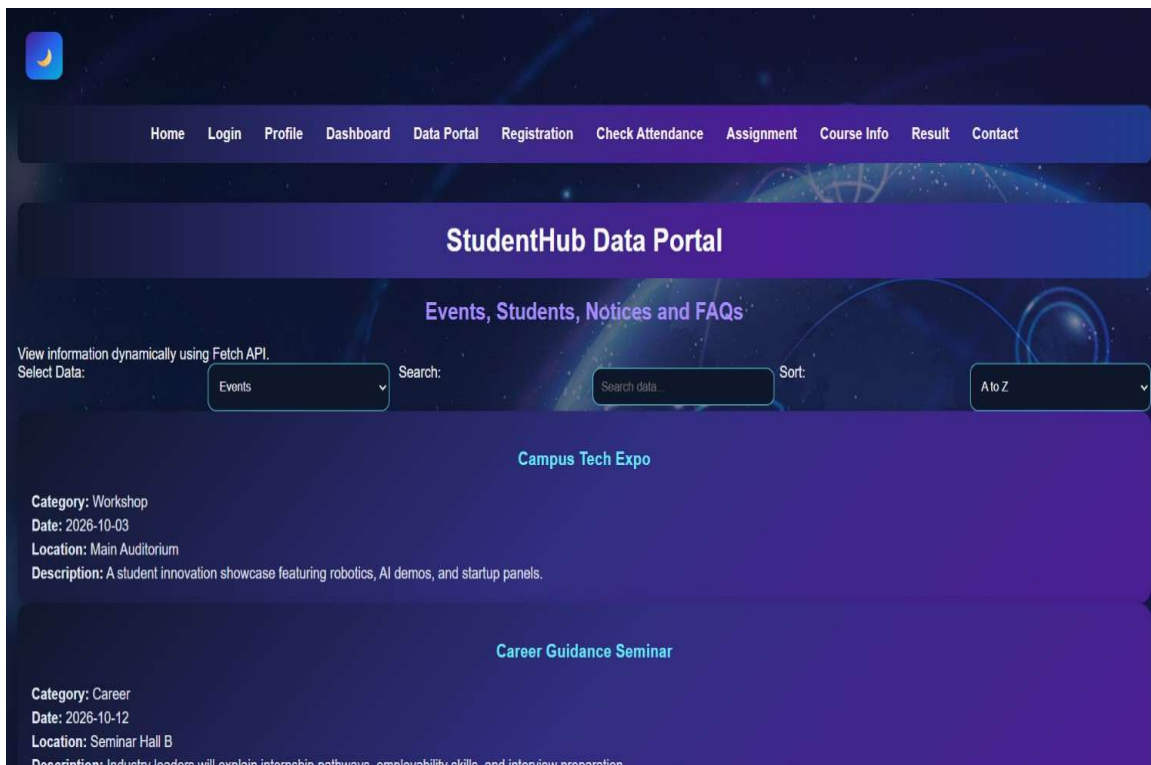
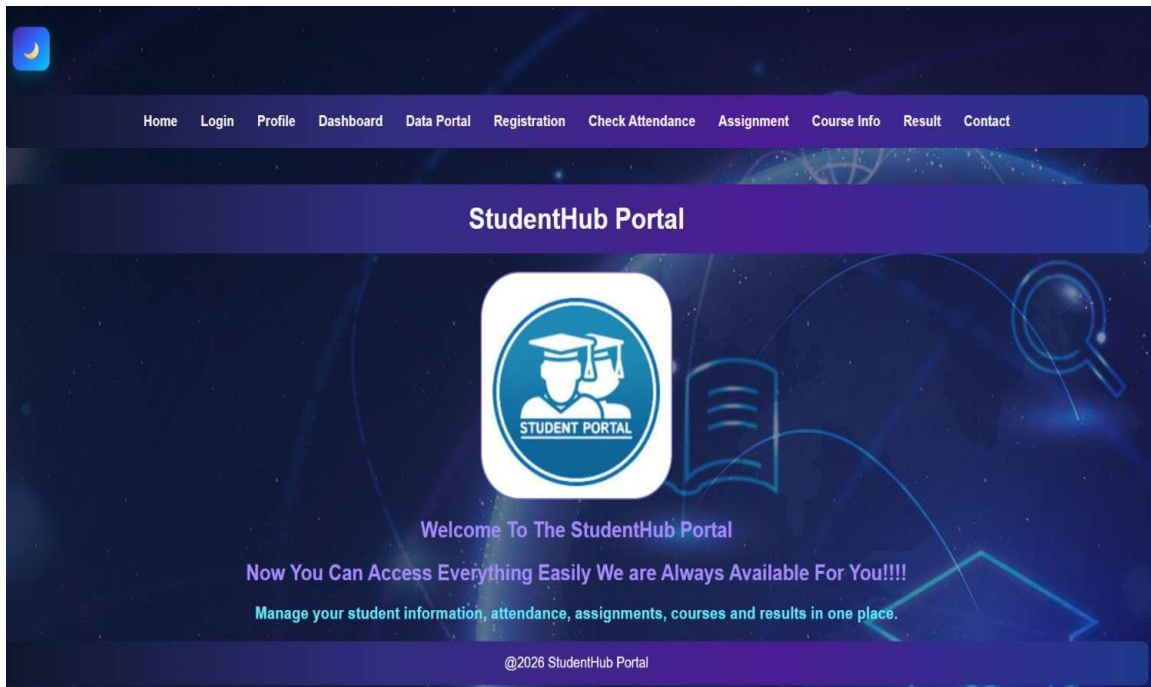
Answer: A loading message is shown while fetching data. If fetching is successful, the data is displayed. If an error occurs, an error message is shown using `catch()`.

4. How is modularity maintained in JavaScript files?

Answer: Modularity is maintained by dividing JavaScript code into separate files and functions, where each file handles a specific task.

Result:

Evidence of Work (Screenshots)



Events

Search data...

A to Z

Campus Tech Expo

Category: Workshop
Date: 2026-10-03
Location: Main Auditorium
Description: A student innovation showcase featuring robotics, AI demos, and startup panels.

Career Guidance Seminar

Category: Career
Date: 2026-10-12
Location: Seminar Hall B
Description: Industry leaders will explain internship pathways, employability skills, and interview preparation.

Cultural Fest Night

Category: Cultural
Date: 2026-10-18
Location: Open Stage
Description: A music, dance, and drama evening with student performances from all departments.

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@2026 StudentHub Portal

StudentHub Data Portal

Events, Students, Notices and FAQs

View information dynamically using Fetch API.
Select Data:

Events

 Search:

Search data...

 Sort:

A to Z

Research Poster Session

Category: Academic
Date: 2026-10-25
Location: Innovation Lab
Description: Undergraduate researchers will present posters on AI, sustainability, and engineering projects.

Sports Championship

Category: Sports
Date: 2026-11-02
Location: College Ground
Description: Interdepartmental tournament featuring cricket, football, and athletics.

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@2026 StudentHub Portal

Events, Students, Notices and FAQs

View information dynamically using Fetch API.

Select Data: Students Search: Sort: Alto Z

Ananya Sharma

Student ID: STU104
 Course: Information Technology
 Year: First Year
 Batch: IT-B
 Email: ananya.sharma@college.edu
 Status: Active

Dev Joshi

Student ID: STU130
 Course: Civil Engineering
 Year: Final Year
 Batch: CIV-D
 Email: dev.joshi@college.edu
 Status: Active

Niket Patel

Student ID: STU101
 Course: Computer Engineering
 Year: Second Year
 Batch: COE-A

Events, Students, Notices and FAQs

View information dynamically using Fetch API.

Select Data: Notices Search: Sort: Alto Z

Fee Payment Deadline

Category: Finance
 Priority: High
 Date: 2026-10-02
 Description: Students who have not paid the tuition fees must complete the payment before 2 October to avoid late charges.

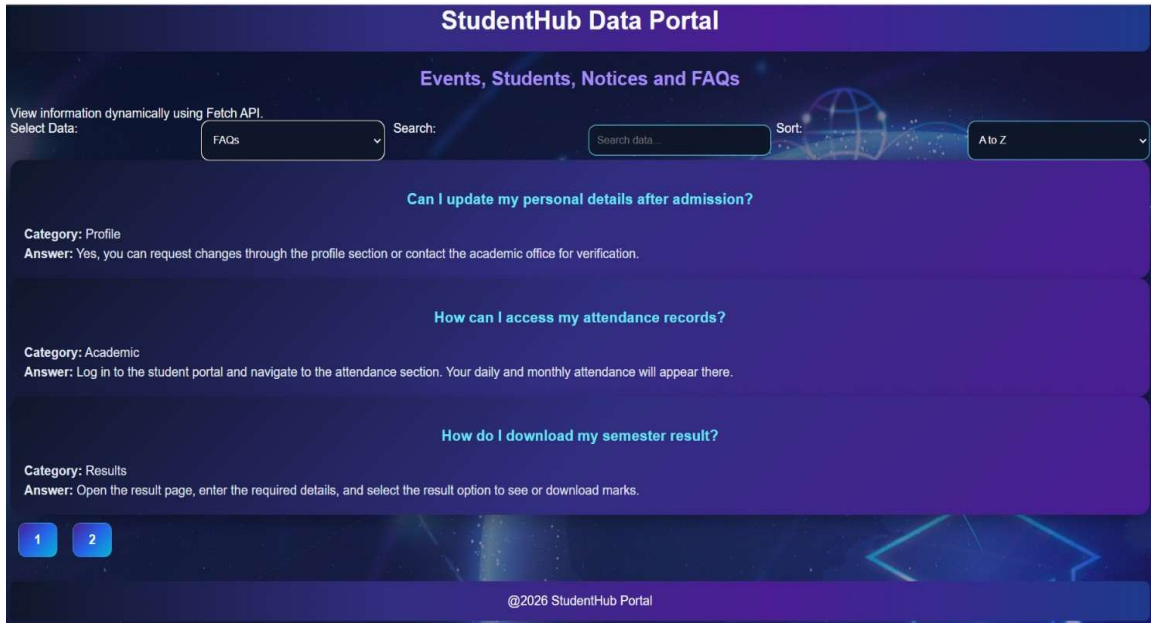
Hostel Maintenance Notice

Category: Campus
 Priority: Low
 Date: 2026-10-10
 Description: Water and sanitation checks will be conducted on 10 October. Residents are advised to cooperate.

Library Hours Extended

Category: Campus
 Priority: Medium
 Date: 2026-09-28
 Description: The central library will remain open until 9:00 PM during revision week for student access.

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A. GitHub Repository Link:

<https://github.com/Niket-Borad/StudentHub-Portal/tree/main/Practical%206>

B. Challenges Faced:

- I faced some difficulty in fetching and reading the JSON data correctly.
- It was a difficult to apply search, filter, and sorting together. like how to update html code and how to apply search operation and fetching.
- I faced problems while implementing pagination for the data.
- Handling loading and error messages was also a challenge.
- Managing the JavaScript code in different files was sometimes confusing. also wrote javascript for this is also difficult .

C. Learning Outcome:-

I learned how to fetch and display JSON data using Fetch API. I also learned search, filter, sorting, and pagination using JavaScript. I understood how to handle loading and error states.

Faculty Remarks: